



Efficacy and safety of efgartigimod in patients with neurological autoimmune diseases

Jin Qin¹, Wei Li, Lipin Yuan, Huiqin Liu, Rui Pang, Jiewen Zhang^{*}

Department of Neurology, Henan Provincial People's Hospital, Zhengzhou University People's Hospital, Zhengzhou, Henan, China

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ABSTRACT

Objective: Myasthenia gravis (MG) is an autoimmune disease characterized by disrupted neuromuscular synaptic transmission. Efgartigimod, a human Fc receptor antagonist, has been approved for patients with MG. Its potential use for other IgG-mediated neurological autoimmune diseases is unclear. This study aimed to retrospectively evaluate the efficacy and safety of efgartigimod in patients with neurological autoimmune diseases. **Methods:** This retrospective study investigated patients with neurological autoimmune diseases who were treated with efgartigimod in the Henan Provincial People's Hospital. The efficacy of the medication was analyzed using the quality-of-life improvement score and the IgG level pre- and post-efgartigimod treatment. The safety of the medication was assessed by considering adverse events and blood parameters. The blood parameters, including routine blood parameters, coagulation, liver, kidney, and immune function. **Results:** Seventeen patients received efgartigimod in the Henan Provincial People's Hospital from September 1, 2023, to January 31, 2024. In MG patients, myasthenia gravis activities of daily living (MG-ADL) reduced after efgartigimod treatment for 4 weeks compared with baseline ($P < 0.05$). Autoimmune encephalitis (AE) is a group of inflammatory disease with antibodies against neuronal synaptic and cell surface antigens. Similarly, patients with AE had a statistically significant reduction in modified Rankin scale (mRS) after efgartigimod treatment for 4 weeks compared with baseline ($P < 0.05$). Guillain Barre syndrome (GBS) is an immune-mediated disease of the peripheral nerves and nerve roots. However, the inflammatory neuropathy cause and treatment (INCAT) scale score didn't statistically differ in GBS patients before and after efgartigimod treatment ($P > 0.05$). The IgG levels significantly reduced after the first infusion and gradually decreased after multiple infusions ($P < 0.05$). Most subjects did not have increased IgG serum levels before treatment. IgA, IgM, and complement levels didn't differ significantly with efgartigimod treatment ($P > 0.05$). There were no changes in blood parameters during the treatment ($P > 0.05$). **Conclusions:** Efgartigimod was effective and safe in neurological IgG-mediated autoimmune diseases, even in patients without increased IgG serum levels.

1. Introduction

Neurological autoimmune diseases are abnormal immune responses against the central nervous system (CNS) and the peripheral nervous system [1]. However, the manifestations of these diseases highly vary, and these manifestations are often severe, resulting in morbidity and mortality [2]. Manifestations of autoimmune diseases result from autoantibody-mediated disorders, T-cell or B-cell mediated disorders, and granulomatous inflammation [3,4]. Neurological autoimmune disorders are commonly treated with immunosuppressive therapies,

including glucocorticoid, intravenous immunoglobulin, plasma exchange (PLEX), and immunoregulators [5]. The standard immunotherapy is often insufficient for some patients to prevent the disease progression.

The neonatal Fc receptor (FcRn) is a major histocompatibility complex (MHC) class I-like molecule that binds to immunoglobulin G (IgG) to prevent lysosomal degradation [6]. It was initially thought that FcRn transports IgG from the mother to the fetus [7]. Subsequently, it was identified that FcRn regulates IgG dynamic levels through distribution, transport, and prevention of degradation [8]. In IgG-mediated

^{*} Corresponding author at: Department of Neurology, Henan Provincial People's Hospital, Zhengzhou 450003, Henan, China.

E-mail address: zhangjiewen9900@126.com (J. Zhang).

¹ First author: Jin Qin

autoimmune diseases, most patients have autoantibodies while a small proportion of patients have no identifiable antibodies. Antagonizing FcRn prevents binding to its target IgG, resulting in the degradation of IgG, including pathogenic autoantibodies with therapeutic effects [9]. Therefore, antagonizing FcRn prevents IgG autoantibodies from binding and results in degradation of antibodies causing IgG-mediated neurological autoimmune diseases.

Efgartigimod is a human IgG1 antibody against Fc fragment with a high affinity to FcRn compared with the endogenous IgG [10]. In phase 1 and 2 trials, efgartigimod significantly reduced the levels of IgG and improved symptoms and activities of daily living in patients with generalized myasthenia gravis [11,12]. Further, efgartigimod was very effective and safe in patients with MG in a phase 3 trial [13]. Recently, several clinical studies of efgartigimod investigated its efficacy for other autoimmune diseases like bullous pemphigoid and immune thrombocytopenia. Efgartigimod significantly increased the meaningful platelet responses compared with placebo in patients with Primary Immune Thrombocytopenia [14,15]. Furthermore, efgartigimod exhibited an early effect on early disease activity for pemphigus [16]. Recently the ADHERE study demonstrated that efgartigimod improved clinical symptoms and decreased the risk of recurrence in patients with chronic inflammatory demyelinating polyradiculoneuropathy (CIDP) compared with placebo [17]. With increasing evidence on its therapeutic effect in MG and CIDP, its efficacy on neurological autoimmune disorders, especially neurological autoimmune disorders without the increased IgG serum levels, remains unclear.

Therefore, this study aimed to assess the efficacy and safety of efgartigimod in neurological autoimmune diseases, especially neurological autoimmune diseases without the increased IgG serum levels. Another objective was to analyze the immune function and adverse reactions in patients with autoimmune neurological diseases following efgartigimod treatment.

2. Methods

2.1. Participants

Patients with neurological autoimmune diseases who received efgartigimod treatment at the Neurology Department of Henan Provincial People's Hospital from September 1, 2023, to January 31, 2024, were included in this analysis. The neurological autoimmune diseases were MG, GBS, AE, neuromyelitis optica spectrum disorder (NMOSD), and polymyositis (PM). NMOSD is an autoimmune disease of the CNS that primarily affects the optic nerves and spinal cord. PM is a kind of idiopathic inflammatory myopathies. Patients with severe infection, acute cardiovascular and cerebrovascular events, and participants who had severe liver and kidney dysfunction were excluded from the study. Among the 18 patients with efgartigimod treatment, 1 case was excluded due to undiagnosed idiopathic inflammatory myopathy, and 17 subjects were subsequently included. All patients consented to efgartigimod therapy and signed informed consent for efgartigimod therapy. This retrospective data collection was in accordance with ethics compliance and relevant regulations. Efgartigimod (10 mg/kg) was intravenously injected at four infusions per cycle (one infusion per week). The efgartigimod regimen was adjusted based on the symptoms' improvement and the patient's economic burden.

2.2. Outcome measures

Clinical parameters were assessed at baseline, 1 week, and 4 weeks after the first infusion of efgartigimod in patients with MG, AE, and GBS using MG-ADL, mRS, and INCAT scale, respectively. Outcomes were defined as significantly favorable with the improvement in MG-ADL ≥ 2 points, in mRS ≥ 1 point and in INCAT ≥ 1 point compared to the baseline, respectively. At the same time, the changes in immune indicators were recorded at baseline and after one infusion, two infusions,

Table 1
A summary of clinical data and outcomes for patients.

#	Diagnosis	Clinical manifestations	Treatments	Outcomes
1	GBS	Inability to stand or hold objects unaided due to limb weakness	Corticosteroids, IVIG, plasma exchange and ofatumumab	Improvement in upper limb weakness but difficulty in walking
2	GBS	Lower limb weakness and limb numbness	Corticosteroids, IVIG and Rituximab	Recovery of motor and sensory function, and ability to walk quickly again
3	GBS	Difficulty in walking and lifting, paresthesia in the limb	Corticosteroids, IVIG, and plasma exchange	Increased muscle power and walk again, accompanied by recovery of numbness
4	AE	Difficulty swallowing and talking, and walk unsteadily	Corticosteroids and MMF	Recovery of swallowing, talking, and walking.
5	AE	Seizures, cognitive impairment and inability to recognize people	No	Reduced seizures, normal communication due to improved cognitive function
6	AE	Seizures and cognitive impairment, poor memory	Corticosteroids	Epilepsy was well controlled and memory showed significant improvement.
7	AE	Behavioral and psychological abnormalities and cognitive impairment	Corticosteroids and IVIG	Recovery of cognitive function and normal life was achieved.
8	AE	Seizures, behavioral and psychological symptoms	Corticosteroids	Reduced seizures and improvement in behavioral abnormalities
9	MG	Eyelid Ptosis, difficulty lifting neck, swallowing and talking	pyridostigmine bromide (Mestinon), corticosteroids, and azathioprine (Imuran)	Significant recovery of neck strength, swallowing and talking. Eyelid Ptosis showed slight improvement
10	MG	Eyelid Ptosis and diplopia	pyridostigmine bromide (Mestinon) and corticosteroids	Disappearance of eyelid Ptosis and diplopia
11	MG	Limb weakness, difficulty in walking, lifting and uncomfortable swallowing	pyridostigmine bromide (Mestinon) and corticosteroids	Restored in limb weakness and ability to walk quickly again, and smooth swallowing.
12	MG	Eyelid Ptosis, weakness in neck and lower limbs	pyridostigmine bromide (Mestinon), corticosteroids and tacrolimus	Improvement in eyes, neck and limb weakness
13	MG	Eyelid Ptosis and dyspnea	pyridostigmine bromide (Mestinon), corticosteroids, IVIG and MMF	Disappearance of dyspnea and eyelid Ptosis showed obvious improvement
14	MG	Eyelid Ptosis	pyridostigmine bromide (Mestinon), corticosteroids and tacrolimus	Eyelid weakness showed obvious improvement.
15	MG	Limb weakness, dyspnea,	pyridostigmine bromide	Improvement in limb weakness,

(continued on next page)

Table 1 (continued)

#	Diagnosis	Clinical manifestations	Treatments	Outcomes
16	MG	difficulty in swallowing and talking Eyelid Ptosis, diplopia, upper limb weakness and difficulty lifting	(Mestinon) and corticosteroids pyridostigmine bromide (Mestinon), corticosteroids and tacrolimus	swallowing and talking and disappearance of dyspnea Restored in eyelid Ptosis and limb weakness returns to normal life.
17	NMOSD	Numbness below the waist and lower limb weakness	Corticosteroids	Improved in paresthesia. Lower limb weakness showed slight improvement.

IVIg: Immunoglobulin, MMF: mycophenolate mofetil (CellCept).

Table 2

Baseline characteristics and medical history of the patients.

	MG (n = 8)	AE (n = 5)	GBS (n = 3)	NMOSD (n = 1)
Age, years	59±14	38±20	23±6	51
Sex				
Male	7 (87.5 %)	3 (60 %)	2 (67 %)	0
Female	1 (12.5 %)	2 (40 %)	1 (33 %)	1 (100 %)
Disease course	4 ± 4 years	0.9 ± 0.7 months	15±19 months	7 years
Previous thymectomy				
Yes	3 (37.5 %)	0	0	0
No	5 (62.5 %)	5 (100 %)	3 (100 %)	1 (100 %)
Combined corticosteroids	8 (100 %)	4 (80 %)	3 (100 %)	1 (100 %)
Infusion of efgartigimod, numbers				
1	1 (12.5 %)	0	1 (33.3 %)	0
2	4 (50 %)	4 (80 %)	0	0
3	1 (12.5 %)	1 (20 %)	0	0
4	2 (25 %)	0	1 (33.3 %)	1 (100 %)
5	0	0	1 (33.3 %)	0

three infusions, and four infusions of efgartigimod. Data are presented as mean ± sd, for data expected to be normally distributed, and as medium (interquartile range) for data expected to not be normally distributed.

The safety of efgartigimod was assessed by the changes in blood parameters and incidence of adverse events. The effects of efgartigimod on blood parameters were assessed during the treatment period. The blood parameters were recorded at baseline and after one infusion, two infusions, three infusions, and four infusions of efgartigimod. Reported adverse events, including headache, respiratory infection, gastrointestinal disorder, sensory abnormality, and muscular pain were evaluated.

2.3. Statistical analysis

SPSS version 23.0 (IBM SPSS, Chicago, IL, USA) was used for statistical analysis. AE The qualitative variables were reported as frequencies. The two groups were compared using the student's *t*-test, while the comparisons of more groups were done using one-way analysis of variance (ANOVA) or Kruskal-Wallis test according to the normality or homogeneity of variance. A *p*-value of < 0.05 was considered statistically significant.

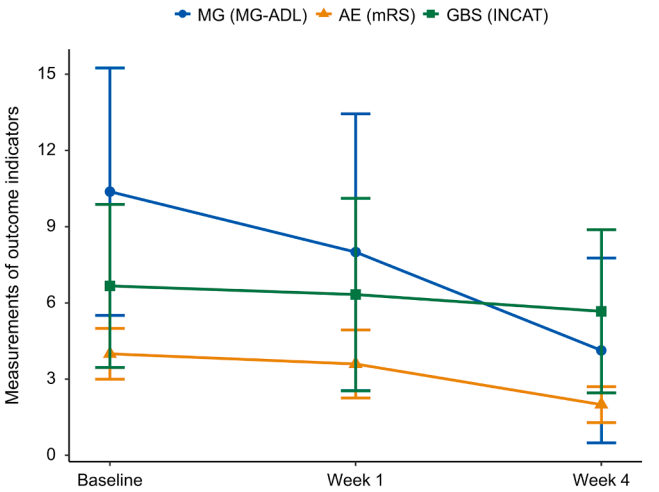


Fig. 1. Change in MG-ADL, mRS, and INCAT pre- and post- efgartigimod therapy.

3. Results

3.1. Baseline characteristics

Seventeen patients, 12 (71 %) males and 5 (29 %) females, who received efgartigimod were investigated. Among them, 8 (47 %) patients had MG, 5 (29 %) patients had AE, 3 (18 %) patients suffered from GBS, and 1 (6 %) patient had NMOSD. We summarized clinical data and outcomes for patients (Table 1). MG patients had 7 (87.5 %) males and 1 (12.5 %) female. AE patients had 3 (60 %) males and 2 (40 %) females. GBS patients had 2 (67 %) males and 1 (33 %) female. The average age of MG patients was 59±14 years and the average course of MG was 4 ± 4 years. The average age of AE patients was 38±20 years and the average course of AE was 0.9 ± 0.7 months. The average age of GBS patients was 23±6 years and the average course of GBS was 15±19 months (Table 2). The patient with NMOSD was a 51-year-old female with a disease course of 7 years.

Based on the Myasthenia Gravis Foundation (MGFA) classification, 2 (25 %) patients had class I MG, 2 (25 %) patients had class IIa MG, 2 (25 %) patients had class IIb MG, and 2 (25 %) patients had class IV MG. Among AE patients, 1 (20 %) patient had anti-N-methyl-D-aspartate-receptor (NMDAR) encephalitis, 3 (60 %) patients had marginal encephalitis, and 1 (20 %) patient had another AE syndrome. There were 1 (33 %) patient displaying signs of acute inflammatory demyelinating polyneuropathy (AIDP) and 2 (67 %) patients exhibiting signs of CIDP. Three (37.5 %) MG patients underwent previous thymectomy. Most patients [16 of 17 (94 %)] received combined hormone therapy. Combined hormone therapy refers to the use of corticosteroids during the treatment of efgartigimod. Two patients had one infusion, 8 patients had two infusions, 2 patients received three infusions, 4 patients received four infusions, and 1 patient received five infusions. MG patients had 1 (12.5 %) patient receiving one infusion, 4 (50 %) patients receiving two infusions, 1 (12.5 %) patient receiving three infusions, and 2 (25 %) patients receiving four infusions. AE patients had 4 (80 %) patients receiving two infusions and 1 (20 %) patient receiving three infusions. GBS patients had 1 patient receiving one infusion, the second patient receiving four infusions, and the third patient receiving five infusions. The NMOSD patient receiving four infusions.

3.2. Clinical characteristics

The MG-ADL score in patients with MG was 10±5 at baseline, 8 ± 5 at 1 week, and 4 ± 4 at 4 weeks after efgartigimod treatment, respectively. Patients with MG showed a statistically significant improvement

Table 3
Outcome measures pre- and post- efgartigimod therapy.

	Baseline	One week after therapy	Four weeks after therapy	p-value
MG-ADL (n = 8)	10 ± 5	8 ± 5	4 ± 4	< 0.05
mRS (n = 5)	4 ± 1	4 ± 1	2 ± 0.7	< 0.05
INCAT (n = 3)	7 ± 3	6 ± 4	6 ± 3	> 0.05

in mean MG-ADL score after 4 weeks of efgartigimod therapy compared with baseline ($P < 0.05$) (Fig. 1) (Table 3).

The mRS in patients with AE was 4 ± 1 at baseline, 4 ± 1 at 1 week, and 2 ± 0.7 at 4 weeks after efgartigimod treatment, respectively. Patients with AE showed a statistically significant improvement in mean mRS after 4 weeks of efgartigimod therapy compared with baseline ($P < 0.05$) (Fig. 1) (Table 3).

The INCAT score in patients with GBS was 7 ± 3 at baseline, 6 ± 4 at 1 week, and 6 ± 3 at 4 weeks after efgartigimod therapy, respectively. Although the INCAT score showed a decreasing trend after 4 weeks of efgartigimod therapy compared with baseline, there was no statistical difference ($P > 0.05$) (Fig. 1) (Table 3).

Twelve (71 %) patients were tested for IgG, IgA, and IgM levels before efgartigimod therapy, without increased IgG serum levels. Compared to the baseline level, the IgG levels were significantly reduced after one infusion ($P < 0.01$), two infusions ($P < 0.05$), and after three infusions ($P < 0.01$). The IgG level gradually decreased after infusions ($P < 0.05$). Nine (53 %) patients were tested for complement 3 (C3) and complement 4 (C4) before efgartigimod therapy, and there were no significant differences in the IgA, IgM, and complement levels after efgartigimod treatment ($P > 0.05$) (Fig. 2) (Table 4).

Each patient's treatment and efficacy were visualized by swimmer plot. The length denotes the duration of treatment and colors denote different diseases. Swimmer plot demonstrated that efgartigimod therapy achieved significant clinical improvement in patients with neurological autoimmune diseases (Fig. 3).

3.3. Safety

No serious adverse events occurred during the efgartigimod therapy. One patient exhibited a mild upper respiratory tract infection. Routine blood parameters at baseline and after one infusion, two, three, and four infusions of efgartigimod treatment were evaluated in 17, 12, 7, 5, and 3 patients, respectively. Blood coagulation at baseline and after one infusion and two infusions of efgartigimod therapy was assessed in 17, 5,

and 3 patients, respectively. Liver function at baseline and after one infusion, two, three, and four infusions of efgartigimod therapy was evaluated in 15, 10, 7, 4, and 3 patients, respectively. Kidney function at baseline and after one infusion, two, three, and four infusions of efgartigimod therapy was assessed in 16, 10, 7, 3, and 3 patients, respectively.

Blood parameters, including routine blood parameters, coagulation time, blood albumin, liver function, and kidney function didn't change with the treatment ($P > 0.05$). Treatment-emergent adverse events (TEAEs), including bleeding, didn't occur (Table 5).

4. Discussion

In this study, the efficacy and safety of efgartigimod for neurological autoimmune diseases were investigated. The study findings showed that efgartigimod was effective and well tolerated in patients with neurological autoimmune diseases, even in patients without increased levels of IgG. Thus, efgartigimod may serve as an optional choice for the treatment of neurological autoimmune diseases.

There are standard immunotherapy treatments for neurological autoimmune diseases. However, it is often insufficient for some patients to prevent the disease progression. Thus, new medications are sought and more attention has been paid to the therapeutic effect of efgartigimod in treating IgG-mediated autoimmune diseases. In recent years, an increasing number of clinical trials on efgartigimod have shown considerable clinical improvement in MG patients [12,13]. In the ADHERE study, efgartigimod improved the symptoms and decreased the risk of recurrence in patients with CIDP. However, information on efgartigimod use in clinical practice is sparse. This study enrolled a broad range of neurological autoimmune diseases, including GBS, AE, and NMOSD. Improvements in quality of life in patients with MG and AE

Table 4
The immune-related parameters pre- and post- efgartigimod therapy.

	Baseline	One week after therapy (n = 9)	Two weeks after therapy (n = 3)	Three weeks after therapy (n = 4)	p-value
IgG, g/L	10 ± 4	6 ± 3	5 ± 2	4 ± 0.9	< 0.05
IgA, g/L	2 ± 0.6	2 ± 1	2 ± 0.9	2 (2, 3)	> 0.05
IgM, g/L	1 ± 0.5	0.9 ± 0.5	0.8 ± 0.1	2 ± 2	> 0.05
C3, g/L	1 ± 0.2	1 ± 0.2	1 ± 0.1	1 (1, 1)	> 0.05
C4, g/L	0.3 ± 0.1	0.3 ± 0.1	0.2 ± 0.9	0.2 (0.2, 0.3)	> 0.05

Abbreviation: C3, Complement 3, C4, Complement 4.

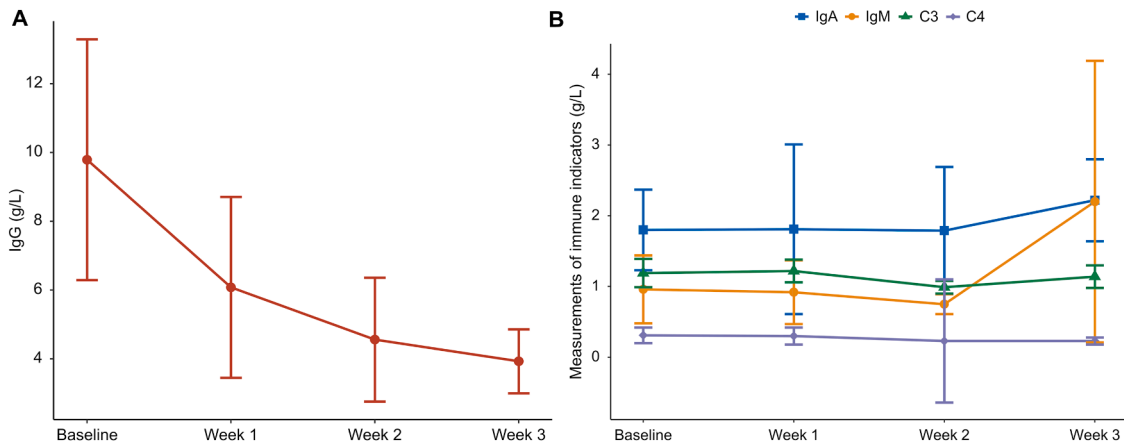


Fig. 2. Change in immune indicators pre- and post- efgartigimod therapy. (A) the levels of IgG, (B) the levels of IgA, IgM, C3 and C4. C3: Complement 3, C4: Complement 4.

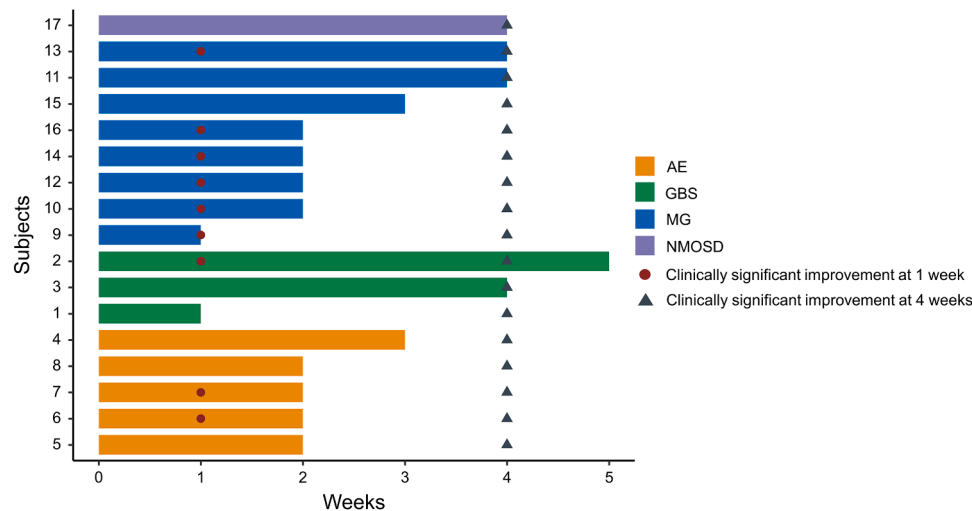


Fig. 3. Patients received efgartigimod were represented in the swimmer plot.

Table 5
Outcome measures for the evaluation of drug safety.

	Baseline	One week after therapy	Two weeks after therapy	Three weeks after therapy	Four weeks after therapy	p-value
WBC, ×10 ⁹ /L	9 ± 2	9 ± 3	9 ± 3	8 (6, 9)	6 ± 1	>0.05
LYC, ×10 ⁹ /L	2 ± 1	2 ± 0.9	2 ± 0.9	2 ± 0.8	2 ± 0.2	>0.05
HGB, g/L	134±16	125±18	134±21	135 (122, 139)	135±12	>0.05
PLT, ×10 ⁹ /L	233±59	226±70	247±36	230±45	244±34	>0.05
PT, s	11 (11, 12)	12±0.7	11±0.4	/	/	>0.05
APTT, s	27±3	30±5	31±4	/	/	>0.05
INR	1 (1, 1.1)	1 ± 0.1	1 ± 0.04	/	/	>0.05
ALT, U/L	19 (15, 26)	31±20	39±24	18 (13, 76)	24±22	>0.05
AST, U/L	19 (15, 23)	18±9	17 (14, 19)	20±15	24±14	>0.05
TP, g/L	61±9	55±6	56±7	56±9	61±3	>0.05
ALB, g/L	38±7	37 (35, 42)	39±6	39±7	43±4	>0.05
TBIL, umol/L	9 (8, 14)	10±4	12±5	10 (8, 10)	7 ± 2	>0.05
CR, umol/L	63±20	52±14	61±17	59±14	59±10	>0.05

Abbreviations: WBC, White blood cell count; C4, Lymphocyte count; HGB, Hemoglobin; PLT, Platelet; PT, Prothrombin time; APTT, activated partial thromboplastin time; INR, international normalized ratio; ALT, alanine aminotransferase; AST, aspartate aminotransferase; TP, Total protein; ALB, Albumin; TBIL, Total bilirubin; CR, Creatinine.

Data are presented as mean ± sd, for data expected to be normally distributed, and as medium (interquartile range) for data expected to not be normally distributed.

were observed, which was consistent with previous research findings in MG patients [12,13]. In patients with GBS, the INCAT score showed a decreasing trend but not with statistical significance. This may have been due to the small number of cases, necessitating more research into other neurological autoimmune diseases.

In previous studies, efgartigimod with or without acetylcholine receptor antibodies reduced IgG levels in MG patients, and the reduction in IgG levels paralleled the improvement in symptoms [12,13]. However, the effect of efgartigimod on neurological autoimmune diseases in patients without increased IgG serum levels remains unclear. In this study, a large portion of patients had normal IgG serum levels before efgartigimod treatment. A significant rapid reduction in total IgG was observed in efgartigimod-treated patients either with normal or increased IgG serum levels at baseline. The reduction was seen after the first administration and continued with each infusion of the medication. Moreover, the improvements in clinical parameters were consistent and long-lasting throughout the 4 weeks of efgartigimod treatment. This outcome may have been due to the clearing of pathogenic IgG antibodies by efgartigimod even in patients without increased IgG serum levels. Moreover, the efgartigimod had no impact on the IgA, IgM, and complement levels, suggesting a selective removal of IgG, possibly pathogenic IgG in neurological autoimmune diseases.

In this study, the incidence of adverse events was low and the adverse events that occurred were generally mild. One patient experienced a

mild upper respiratory tract infection during the efgartigimod treatment. Consistent with the literature results, efgartigimod was well tolerated in this study [12,13]. Although the IgG reduction by efgartigimod was observed, there was no impact on albumin levels, which was consistent with previous research [13]. This study comprehensively evaluated the safety of efgartigimod. The clinical laboratory parameters, including blood cells, coagulation, blood albumin, liver function, and kidney function were not noticeably changed. Thus, efgartigimod was well tolerated in patients with neurological autoimmune diseases.

This retrospective analysis had several limitations. Unlike prospective studies, this study had a lack of standardized protocols for efgartigimod, insufficiency in evaluating therapeutic effects, and difficulty in determining the economic burden during the treatment. Some patients didn't complete a cycle with efgartigimod. Secondly, weekly scores on the MG-ADL, INCAT, or mRS scores were not obtained in patients with autoimmune diseases from the retrospectively collected data. Moreover, not all patients were not tested for the IgG levels before efgartigimod treatment. Finally, blood parameters were only obtained in a subset of patients and these outcome measures were not prospectively collected.

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CRediT authorship contribution statement

Jin Qin: Writing – original draft, Investigation, Formal analysis, Conceptualization. **Wei Li:** Writing – original draft, Investigation. **Lipin Yuan:** Investigation, Data curation. **Huiqin Liu:** Formal analysis, Data curation. **Rui Pang:** Formal analysis. **Jiwen Zhang:** Writing – review & editing, Supervision, Project administration, Conceptualization.

Declaration of competing interest

We declare that all authors have no competing conflicts of interest.

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